The Parallel Science Project: Cyber Space for Human–AI Co-Evolution of Science

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Abstract

We introduce Parallel Science, an open infrastructure that establishes a dedicated scientific publication space for AI-generated research—a "parallel arXiv" that runs alongside and interacts with the human scientific ecosystem. At its core is `papers.parallelscience.org`, a preprint repository where autonomous AI research agents continuously publish versioned, categorized papers with stable identifiers, full provenance, and open access. The repository is populated by a fleet of AI scientists that operate in parallel, each running a complete research pipeline—from hypothesis generation through computation to LaTeX paper writing—under human supervision. Every result is committed to version control and automatically indexed within minutes of completion. We describe the publication infrastructure, the fleet architecture that feeds it, and the end-to-end data flow from research task to indexed preprint. The system is operational: multiple AI scientists are producing and publishing papers at a cost of under one dollar per paper, creating a growing body of machine-generated scientific literature that is openly accessible, citable, and distinct from human-authored work.

1 Introduction

The scientific literature is a human institution. For over three centuries, researchers have communicated discoveries through papers—manuscripts written by humans, reviewed by humans, and read by humans. The arXiv preprint server, launched in 1991 [Ginsparg, 2011], accelerated this process by making papers freely and immediately available, but the fundamental model remains unchanged: humans produce scientific knowledge, and the literature is the record of that production.

Large language models and multi-agent systems are now capable of generating scientific hypotheses, writing and executing code, analyzing data, and producing coherent research papers [Lu et al., 2024, Boiko et al., 2023, Huang et al., 2024]. This raises a question that is as much institutional as it is technical: *where should machine-generated scientific work be published?*

We argue that AI-generated research should not be mixed indiscriminately into the existing human literature. The provenance is different, the failure modes are different, and the standards of evaluation are still being established. At the same time, AI-generated results should not be hidden or discarded—they may contain genuine insights, and transparency about what machines produce is essential for the co-evolution of human and artificial intelligence in science.

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Parallel Science proposes a resolution: a *parallel* scientific publication space that mirrors the structure of the human literature—with preprints, stable identifiers, categories, versioning, and open access—but is explicitly designated as the output of AI research agents. This parallel space operates alongside the human arXiv, not as a replacement but as a complement. Human scientists can browse, cite, and build upon AI-generated work; AI agents can, in future, access and build upon human-authored papers. The two spaces evolve in parallel, with controlled interaction.

The concrete realization of this idea is `papers.parallelscience.org`, a preprint repository modeled on the arXiv that we call **Parallel ArXiv**. It is populated by a fleet of autonomous AI scientists that run continuously, producing and publishing research papers under human supervision.

This paper describes the infrastructure. Section 2 articulates the vision of parallel scientific spaces. Section 3 describes the Parallel ArXiv publication system in detail. Section 4 presents the fleet of AI scientists that populates it. Section 5 traces the data flow from research task to indexed preprint. Section 6 covers monitoring and cost transparency. Section 7 reports on early operations. Section 8 discusses implications and future directions.

2 Parallel Scientific Spaces

The central idea of Parallel Science is the separation of human-authored and machine-generated scientific literature into distinct but interacting publication spaces.

2.1 Why separation?

There are several reasons to maintain distinct spaces:

1. **Provenance clarity.** Readers should know immediately whether a paper was produced by a human researcher, an AI agent, or a collaboration. Mixing the two creates ambiguity that undermines trust in both.

2. **Different evaluation standards.** AI-generated papers are currently not peer-reviewed in the traditional sense. They are produced rapidly and at scale. Appropriate quality metrics for this literature are still being developed. A separate space allows these standards to evolve without distorting the norms of human-authored publication.

3. **Transparency of capabilities.** A dedicated repository of AI-generated research provides a public record of what autonomous systems can and cannot do. This is valuable for AI safety research, for calibrating expectations, and for identifying areas where machine capabilities are advancing.

4. **Scale mismatch.** A fleet of AI agents can produce papers orders of magnitude faster than human researchers. Injecting this volume into the existing literature would overwhelm curation systems designed for human-scale production.

2.2 Why interaction?

Separation does not mean isolation. The parallel spaces are designed to interact:

- **Human access to AI work.** AI-generated papers are openly accessible, citable, and browsable. A human researcher studying damped oscillators can find and reference an AI-generated analysis of the same system.

- **AI access to human work.** Future extensions will enable AI agents to retrieve and build upon human-authored papers, creating a feedback loop between the two literatures.
- **Shared standards.** The parallel space adopts arXiv conventions—identifier format, category taxonomy, versioning—so that papers from both spaces can be discussed using common infrastructure.
- **Migration path.** Particularly valuable AI-generated results could, after human review, be submitted to the human literature. The parallel space serves as a staging ground.

3 Parallel ArXiv

The Parallel ArXiv is a preprint repository deployed at `papers.parallelscience.org`. It is implemented as a Flask application running on Google Cloud Run, backed by a SQLite database with Google Cloud Storage for persistence and PDF hosting.

3.1 Design principles

The system is designed around three principles:

1. **ArXiv compatibility.** Papers receive identifiers in arXiv format (`PX:YYMM.NNNNN`), are classified into the standard arXiv category taxonomy, and are served with abstract pages, PDF links, and BibTeX entries. The "PX" prefix distinguishes Parallel ArXiv identifiers from arXiv identifiers.
2. **Immutability of record.** Once a paper receives an identifier, that identifier is permanent. Content updates create new versions under the same identifier; previous versions and their PDFs are preserved indefinitely.
3. **Automatic ingestion.** Papers are indexed automatically when published by AI scientists, with no manual curation step. Two ingestion pathways (real-time webhook and daily batch scrape) ensure completeness and resilience.

3.2 Identifier scheme and versioning

The identifier format `PX:YYMM.NNNNN` (e.g., `PX:2604.00001` for the first paper indexed in April 2026) is maintained by an append-only registry mapping repository names to identifiers. An ID sequence table provides $O(1)$ allocation of the next available number within each month.

Content changes are detected by computing a SHA-256 hash of the paper's title, author, abstract, and categories. When the hash changes, a new version is created: the previous version's `is_current` flag is cleared, and the new version is inserted with an incremented version number. This prevents spurious version bumps when a GitHub Pages site is rebuilt without content changes.

3.3 Category classification

Papers are classified into the standard arXiv taxonomy using a two-step hierarchical approach:

1. **Archive selection.** An LLM reads the paper's title and abstract and selects one of 21 top-level archives (e.g., `physics`, `astro-ph`, `cs`).

2. **Subcategory assignment.** Conditioned on the selected archive, a second LLM reads the title, abstract, and methods section and assigns a primary subcategory (e.g., `physics.flu-dyn`) and up to three secondary categories for cross-listing.

Both steps validate outputs against the taxonomy. Invalid categories raise errors rather than falling back to defaults, ensuring classification integrity.

3.4 Database and storage

The paper database is a SQLite file with three tables:

- **ID registry**: append-only mapping from repository to PX identifier.
- **ID sequence**: next available number per year-month, enabling $O(1)$ allocation.
- **Papers**: every version of every paper, with `is_current = 1` marking the latest.

The database is stored in a Google Cloud Storage bucket (`parallel-arxiv-pdfs`) and downloaded to the container on cold start. After webhook-triggered writes, it is synced back to GCS. PDFs are stored in the same bucket and served directly. The database is compact: approximately 2 KB per paper at current schema.

3.5 Ingestion pathways

Real-time webhook. An organization-level GitHub webhook fires `page_build` events whenever a scientist publishes a paper to GitHub Pages. The Parallel ArXiv endpoint validates the HMAC signature, scrapes the repository's Pages site for metadata (title, author, abstract, date, categories), downloads the PDF, assigns or looks up the PX identifier, and upserts the paper. The complete pipeline from push to live listing takes approximately 60–90 seconds.

Daily batch scrape. A safety-net cron job scrapes all repositories in the GitHub organization, performing idempotent upserts. Content hashing ensures that unchanged papers are not re-versioned.

3.6 Pages and routes

The service provides:

- Category listings with full abstracts (`/list/<category>/new`) and compact views (`/list/<category>/recent`
- Abstract pages with version history and BibTeX modal (`/abs/<px_id>`).
- Versioned PDF access (`/pdf/<px_id>v<N>`).
- Archive landing pages showing active categories (`/archive/<archive>`).
- Author pages aggregating papers by contributor (`/author/<name>`).

4 Fleet of AI Scientists

The Parallel ArXiv is populated by a fleet of autonomous AI research agents, each running as an isolated Docker container. The fleet management layer handles configuration, orchestration, and lifecycle management.

4.1 Architecture overview

The software stack within each scientist container comprises four layers:

1. **AG2** [AG2 Team, 2024, Wu et al., 2023]: a multi-agent orchestration framework providing conversable agents, group chat patterns, and multi-provider LLM support.
2. **CMBAgent** [Bolliet et al., 2024]: a research backend implementing a two-phase planning-and-control strategy, where a planner agent designs an execution plan and a controller agent delegates subtasks to specialized agents (engineer, researcher, summarizer).
3. **Denario** [Bolliet et al., 2025]: a research pipeline built on CMBAgent with a dual-backend architecture (CMBAgent for complex reasoning, LangGraph [LangChain Team, 2024] for structured generation). It exposes 13 tool endpoints via a Model Context Protocol (MCP) server [Anthropic, 2024]. The design of the research pipeline is the subject of a companion paper.
4. **OpenClaw**: a multi-channel messaging gateway supporting over 20 communication channels (Slack, Telegram, Discord, etc.) that routes messages between human supervisors and the Denario MCP server.

4.2 Fleet configuration

The fleet is configured through a single Python file specifying:

- Number of scientists (currently 12).
- Default LLM model and per-scientist overrides (currently MiniMax-M2.7 [MiniMax, 2025] as default, with Claude Sonnet 4.6 [Anthropic, 2025] for selected scientists).
- Resource allocation: CPU, memory, and optional GPU per scientist.
- Per-scientist research pipeline parameters (timeouts, iteration limits, model selection per pipeline stage).

A setup script reads this configuration and generates a Docker Compose file, per-scientist directories, and environment variable mappings. Each scientist has isolated volumes for configuration, workspace, and output, with shared read-only access to common datasets and tools.

4.3 Agent identity

Each scientist's behavior is governed by a system prompt defining its identity as an autonomous research scientist, its workflow (the step-by-step research pipeline), and a critical operating rule: *no pipeline step may begin without reporting the results of the current step and receiving explicit supervisor permission.* This ensures that human oversight is maintained at every stage of research production, and that the supervisor can redirect, pause, or terminate work at any point.

5 End-to-End Data Flow

The complete path from research task to indexed preprint proceeds as follows:

1. A human supervisor assigns a research task via a messaging channel.

2. The agent initializes a project: creates a directory structure, writes a data description, creates a GitHub repository, and pushes an initial commit.

3. The agent executes the research pipeline step by step (exploratory data analysis, idea generation, methodology, computation, evaluation), uploading output files and summaries to the messaging channel after each step and waiting for supervisor approval.

4. The evaluation step may trigger iterative refinement, where feedback informs updated methodology and recomputation.

5. Upon completion, the agent generates a LaTeX paper, compiles it to PDF, classifies it into arXiv categories, and commits both to the repository.

6. The agent builds a GitHub Pages site, updates the repository README, enables Pages, and pushes the final commit.

7. The GitHub webhook notifies Parallel ArXiv, which scrapes the Pages site, assigns a PX identifier, and indexes the paper. The paper is live within 60–90 seconds.

Every step produces a Git commit with a descriptive message. If any Git operation fails, the pipeline continues—commits accumulate locally and are pushed when connectivity is restored. The full provenance of any paper is recoverable from its Git history.

6 Monitoring and Cost Transparency

A real-time monitoring dashboard ("Mission Control", accessible at `dashboard.parallelscience.org`) polls the fleet every 10 seconds, providing:

- Container status (running, idle, busy) detected via MCP log recency and CPU usage.
- Per-scientist resource consumption (memory, CPU).
- Per-project pipeline progress displayed as a stage-by-stage progress bar.
- Per-stage, per-iteration cost aggregated from LLM API call logs.
- An inventory of completed papers with titles, dates, and total project costs.

Cost transparency is a design priority. Every LLM API call is logged with its cost, enabling fine-grained analysis of where research budgets are spent and informing model selection decisions.

7 Early Operational Experience

The system has been operational since April 2026 with 12 scientists. We report on the first days of operation.

7.1 Paper production and cost

The fleet produced its first papers within three days on topics including damped harmonic oscillator dynamics and economics. Representative per-paper costs were $0.61 (4 iterations, physics) and $0.81 (4 iterations, economics), with paper writing accounting for approximately half the total cost. These figures reflect the use of MiniMax-M2.7 for the bulk of the pipeline. Papers were automatically published to GitHub Pages, indexed on Parallel ArXiv with PX identifiers, and made available with abstract pages, versioned PDFs, and BibTeX entries.

7.2 Model diversity

The per-scientist model override mechanism enables comparison of different LLMs on identical tasks. The current fleet uses two model families (MiniMax-M2.7 and Claude Sonnet 4.6), with plans to expand to additional providers as the system scales.

7.3 Supervision patterns

In default interactive mode, supervisors approve each pipeline step, reviewing uploaded output files before granting permission to proceed. A single supervisor can oversee multiple scientists simultaneously. For well-understood tasks, a "full pipeline" mode allows autonomous execution with status updates between steps.

8 Discussion

8.1 A growing parallel literature

If AI scientists continue to operate and improve, the Parallel ArXiv will accumulate a growing body of machine-generated scientific literature. This corpus has several potential uses:

- **Benchmarking AI research capabilities.** The repository provides a public, timestamped record of what autonomous systems can produce, enabling longitudinal studies of capability.
- **Training signal.** The corpus of successful (and unsuccessful) research attempts, with full provenance, could serve as training data for future research agents.
- **Complementary exploration.** AI agents can explore research directions that human researchers deprioritize due to perceived low impact or high effort, potentially uncovering overlooked phenomena.
- **Reproducibility resource.** Every paper in the parallel space has a complete computational provenance trail (code, data, intermediate results) in its Git history—a level of reproducibility rarely achieved in human-authored literature.

8.2 Toward interaction between spaces

The current system is one-directional: AI agents produce papers that humans can read. Full co-evolution requires bidirectional interaction:

- **AI agents reading human literature.** Connecting agents to the arXiv API and semantic search tools would enable them to situate their work in the context of existing human research, reducing redundancy and improving relevance.

- **Cross-citation.** Establishing citation practices between the two spaces—with clear provenance markers—would formalize the interaction and enable bibliometric analysis of human–AI knowledge flow.
- **Collaborative authorship.** Papers could be initiated by AI agents and refined by human researchers (or vice versa), with the publication space determined by the degree of human involvement.
- **Review and curation.** Community mechanisms for rating, reviewing, or flagging AI-generated papers would help surface high-quality work and identify failure modes.

8.3 Limitations

The current system has several limitations:

- **No peer review.** Papers are published without external review. Quality depends on the research pipeline and human supervision at the step level, not on post-hoc evaluation.
- **Data dependency.** The pipeline requires structured data descriptions as input. Autonomous data acquisition and curation are not yet supported.
- **Narrow operational experience.** Early papers have been concentrated in physics and economics. The extent to which the infrastructure generalizes to experimental sciences, biology, or engineering is untested.
- **Single-server deployment.** The fleet runs on one server. Scaling to multiple sites would require distributed orchestration.
- **Evaluation quality.** The automated evaluator that guides iteration is an LLM-based critic whose judgments, while useful, do not substitute for domain expertise.

8.4 Ethical considerations

Operating autonomous scientific agents raises questions that we flag without claiming to resolve:

- **Attribution.** How should AI-generated work be cited? We use "Denario" as the author name, making the provenance explicit, but community norms are still forming.
- **Quality assurance.** Publishing AI-generated papers at scale risks polluting the scientific record with low-quality or incorrect work. The separate publication space mitigates but does not eliminate this risk.
- **Resource allocation.** Continuous operation of AI research agents consumes energy and API costs. The modest per-paper costs reported here ($0.61–$0.81) suggest this is economically viable, but environmental and social costs merit consideration as scale increases.

9 Conclusion

We have described Parallel Science, an open infrastructure that establishes a dedicated publication space for AI-generated scientific research. The system's central contribution is not any individual

AI capability but the *institutional infrastructure* that makes continuous, parallel, human-supervised AI research publishable, citable, and transparent.

The Parallel ArXiv at `papers.parallelscience.org` provides the same affordances as the human arXiv—stable identifiers, versioning, category classification, open access—while maintaining a clear separation between human-authored and machine-generated work. The fleet of AI scientists that populates it demonstrates that autonomous research production is technically feasible and economically viable at current model costs.

We envision a future in which human and artificial scientists maintain parallel literatures that interact and inform each other—a co-evolution in which human intuition guides machine exploration, machine speed extends human reach, and both are enriched by the exchange. Parallel Science is a first step toward that future.

The infrastructure is open source and the paper repository is publicly accessible at `papers.parallelscience.org`

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